

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with sub-agent orchestration and human-in-the-loop validation (Context Researcher → SQL Drafter → Validator → Devil's Advocate); 8 deterministic SQL rules and a 23-case golden-query evaluation suite. Partnered with Product, Data Science, and business stakeholders to translate ambiguous business questions into agent-ready specs. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures — eliminated the analyst as the bottleneck for routine asks.
- Built a registry-governed semantic layer over BigQuery that exposes business meaning to natural-language queries: YAML definitions plus SQLite vocabulary snapshots define legal dimensions, values, and table wiring; a deterministic prompt builder generates the LLM contract from the registry; a single LLM call maps the question to strict JSON intent (the model never writes SQL); a deterministic gate validates and canonicalizes intent into a typed dataclass before a deterministic compiler emits the SQL — same question, same answer, every time. New or changed definitions pass a human review gate before entering the registry, making the only probabilistic step regression-testable.
- Built an autonomous Jira ticket-resolution agent: 6-gate triage, MCP-driven context gathering, FAISS+BM25 retrieval over 11,000 historical SQL queries, self-healing execution with a 10-retry budget. Closed 400+ data requests across 8 categories over 15 months — cut routine turnaround from 30–60 minutes per request (and up to a full day for complex ones) to under 10 minutes each. Stakeholders never knew an agent was answering them.
- Replaced ~20 distinct recruitment workflows I'd been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform. Stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates. A recurring 3-hour task now takes ~3 minutes of setup.
- Built a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; state persists to BigQuery so the agent has no LLM-memory reliance. Designed a v2 architecture as a 3-tier event-driven platform (reactive events, threshold KPIs, PM-owned themes) self-configuring via a BigQuery event registry. New event types ship without code deploys.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- Client-facing RAG-to-proposal generator deployed on AWS EC2; cut proposal time ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Data & SQL: BigQuery, SQL, dbt, Python, PySpark, Airflow, Databricks, MySQL

Semantic modeling / analytics engineering: registry-governed semantic layer (YAML definitions + SQLite vocabulary snapshots), deterministic SQL compiler, strict-JSON intent contracts, value canonicalization, evaluation suites (golden sets, human-in-the-loop validation)

AI-assisted development: Claude Code, Cursor (daily primary dev environment)

Agent stack: MCP (Model Context Protocol) server design, sub-agent orchestration, context engineering, prompt engineering, RAG (FAISS + BM25 + RRF), multi-model routing across providers, headless cron agents, event-driven architectures, decision logging / observability

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

APIs / triggers / glue: FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores, CI/CD (pytest, version control)

Viz: Power BI (DAX, XMLA TMSL), Tableau, LookML, Streamlit, React Recharts

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)	<i>Chicago, IL</i>
Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)	<i>Chicago, IL</i>